

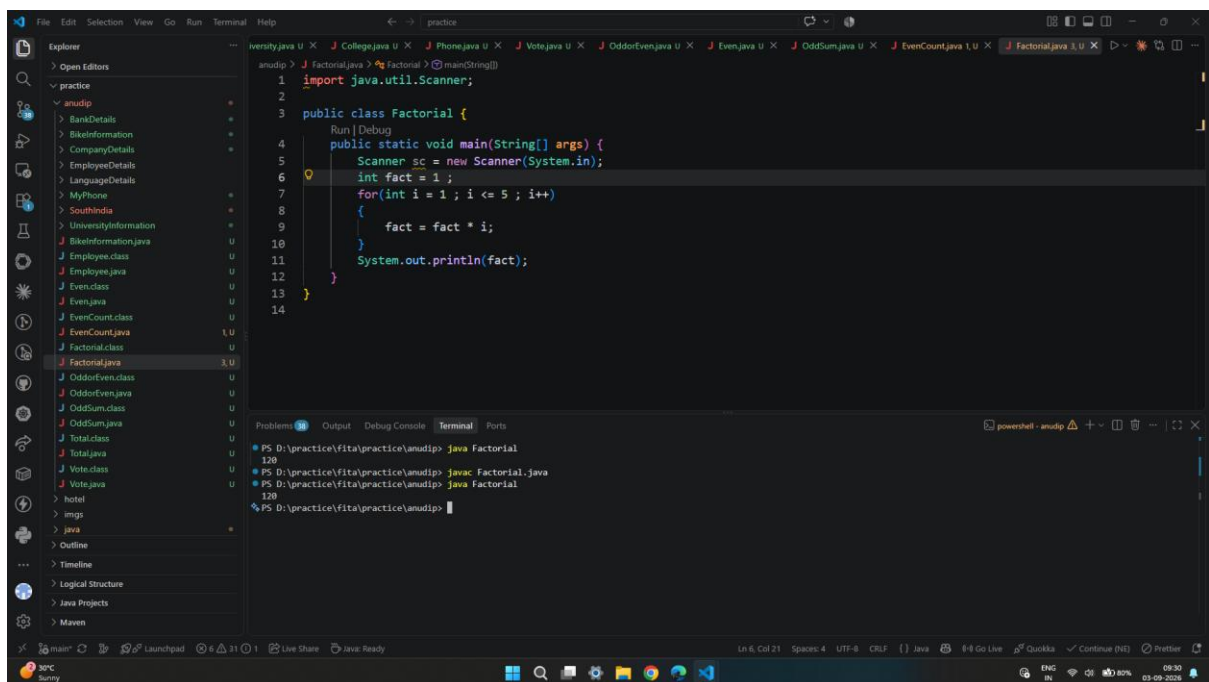
QUESTION 6:

Description: Write a program to find the factorial of a number.

Example:

Input = 5

Output = 120



The screenshot shows an IDE with a Java project named 'practice'. The 'Factoial.java' file is open, containing the following code:

```
1 import java.util.Scanner;
2
3 public class Factorial {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         int fact = 1;
7         for(int i = 1; i <= 5; i++)
8         {
9             fact = fact * i;
10        }
11        System.out.println(fact);
12    }
13 }
14
```

The terminal output shows the execution of the program:

```
PS D:\practice\fta\practice\anudip> java Factorial
120
PS D:\practice\fta\practice\anudip> javac Factorial.java
PS D:\practice\fta\practice\anudip> java Factorial
120
PS D:\practice\fta\practice\anudip>
```

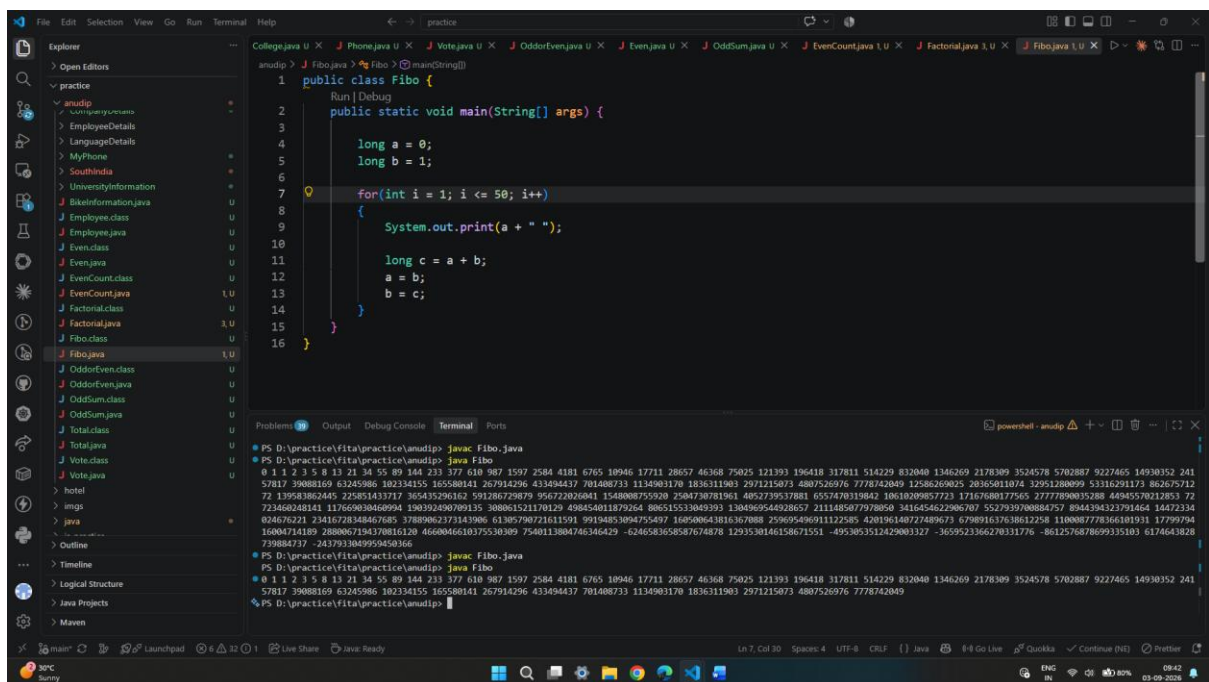
The status bar at the bottom indicates the file is at Line 6, Column 21, with 4 spaces, UTF-8 encoding, and CRLF line endings. The IDE is running on a Windows 10 system with a taskbar showing various applications and the system clock at 09:30 on 03-09-2020.

QUESTION 7:

Description: Write a program to print the fibonacci series of a number 1 to 100.

Example:

Output = 0,1,1,2,3,5.....



```
1 public class Fibo {
2     public static void main(String[] args) {
3
4         long a = 0;
5         long b = 1;
6
7         for(int i = 1; i <= 50; i++)
8         {
9             System.out.print(a + " ");
10
11             long c = a + b;
12             a = b;
13             b = c;
14         }
15     }
16 }
```

PS D:\practice\practise\anudip> javac Fibo.java

PS D:\practice\practise\anudip> java Fibo

0 1 1 2 3 5 8 13 21 34 55 89 144 233 377 610 987 1597 2584 4181 6765 10946 17711 28657 46368 75025 121393 196418 317811 514229 832040 1346269 2178309 3524578 5702887 9227465 14930352 24157817 39988169 63245986 102334155 165580141 267914296 433494437 701408733 1134903170 1836311903 2971215073 4807526976 7787420849 12586269025 20365011074 32951280099 53162911173 86267571272 139583862445 225851433717 365435296162 591286728679 956722026041 1548088755920 2504738781961 4062739937881 6551478315842 1061030967723 17167688177565 27777890833288 44945570222853 723468048141 11766903046994 190392490709135 308061521170129 498454011879264 806515533049393 1304969544928657 2111485877978050 3416454622906787 5527939700884757 8944394323791464 14472334024676221 23416728348467685 37889062373143906 61305790721611591 99194853094755497 160500643816367088 259695496911122585 420196140727489673 679891637638612258 1100087778366101931 1779979416084714189 2880067194370816120 466004610375530389 7540113804746346429 -6246583658587674878 1293530146158671551 -4953053512429003327 -3659523366270331776 -8612576878699335183 6174643828795884737 -243793304995940346

PS D:\practice\practise\anudip> javac Fibo.java

PS D:\practice\practise\anudip> java Fibo

0 1 1 2 3 5 8 13 21 34 55 89 144 233 377 610 987 1597 2584 4181 6765 10946 17711 28657 46368 75025 121393 196418 317811 514229 832040 1346269 2178309 3524578 5702887 9227465 14930352 24157817 39988169 63245986 102334155 165580141 267914296 433494437 701408733 1134903170 1836311903 2971215073 4807526976 7787420849

PS D:\practice\practise\anudip>

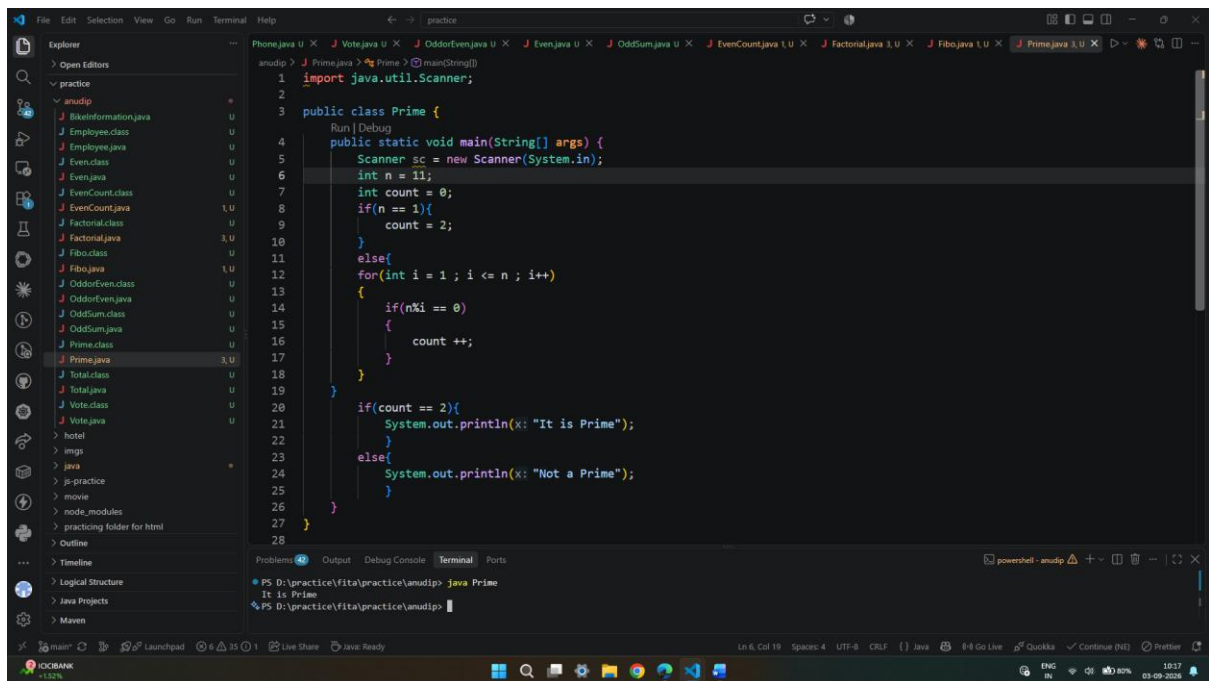
QUESTION 8:

Description: Find prime number or not.

Example:

Input = 11

Output = prime number



```
1 import java.util.Scanner;
2
3 public class Prime {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         int n = 11;
7         int count = 0;
8         if(n == 1){
9             count = 2;
10        }
11        else{
12            for(int i = 1 ; i <= n ; i++){
13                if(n%i == 0)
14                {
15                    count ++;
16                }
17            }
18        }
19        if(count == 2){
20            System.out.println(x: "It is Prime");
21        }
22        else{
23            System.out.println(x: "Not a Prime");
24        }
25    }
26 }
27
28
```

PS D:\practice\fta\practice\anudip> java Prime
It is Prime
PS D:\practice\fta\practice\anudip>

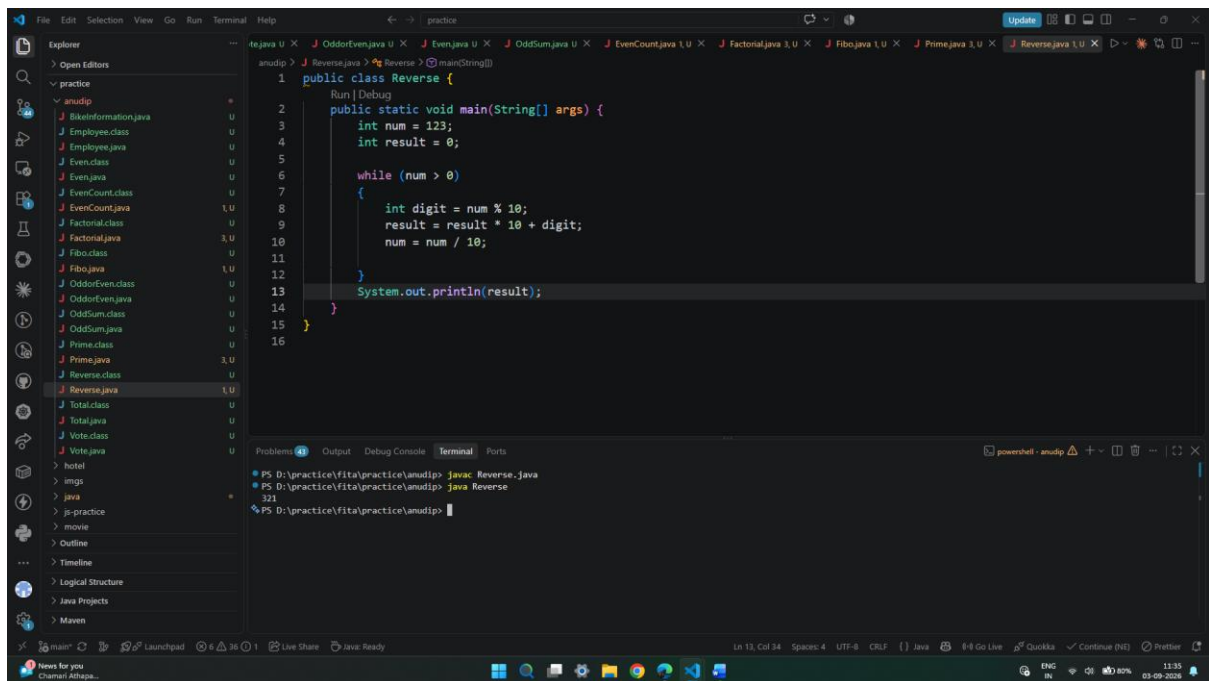
QUESTION 9:

Description: Reverse the number

Example:

Input = 123

Output = 321



The screenshot shows an IDE with a project named 'practice'. The Explorer panel on the left lists various Java files, including 'Reverse.java'. The main editor displays the code for 'Reverse.java':

```
1 public class Reverse {  
2     public static void main(String[] args) {  
3         int num = 123;  
4         int result = 0;  
5  
6         while (num > 0)  
7         {  
8             int digit = num % 10;  
9             result = result * 10 + digit;  
10            num = num / 10;  
11        }  
12        System.out.println(result);  
13    }  
14 }  
15  
16 }
```

The bottom panel shows the 'Terminal' output:

```
PS D:\practice\fta\practice\anudip> javac Reverse.java  
PS D:\practice\fta\practice\anudip> java Reverse  
321  
PS D:\practice\fta\practice\anudip>
```

The status bar at the bottom indicates the file is 'Ln 13, Col 34', uses 'UTF-8' encoding, and is a 'Java' file.

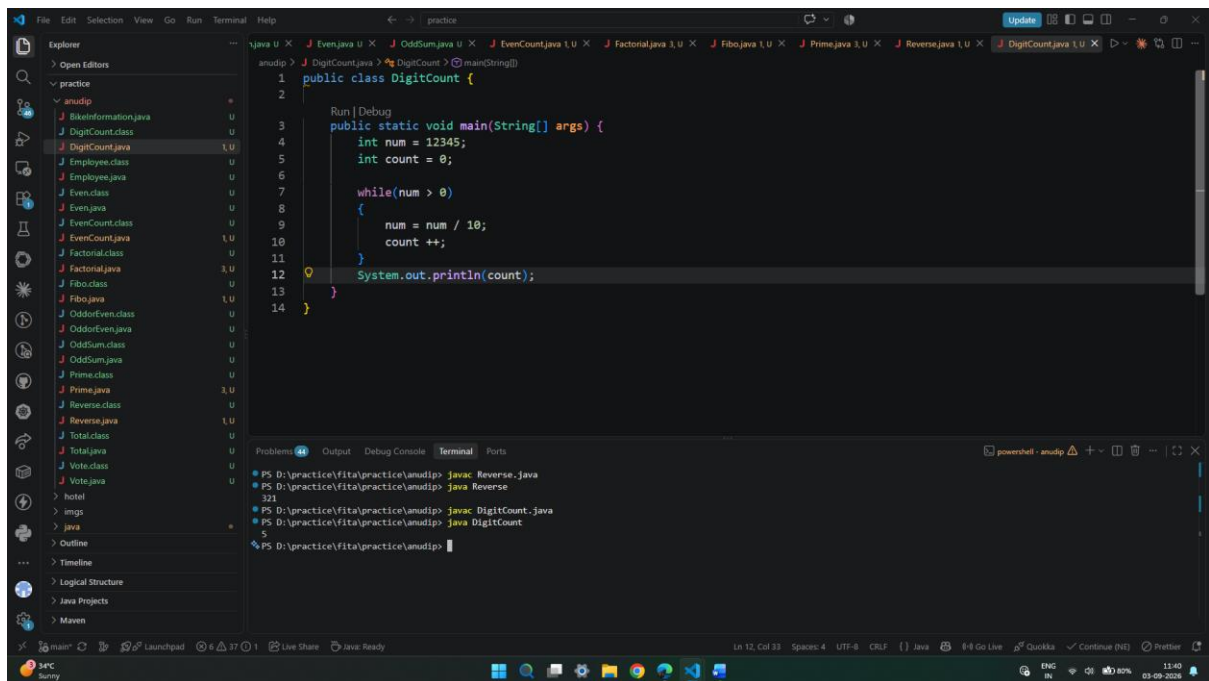
QUESTION 10:

Description: Count of the number

Example:

Input = 123

Output = 3



The screenshot shows a code editor with a Java program named `DigitCount.java`. The program defines a `DigitCount` class with a `main` method that takes an array of strings `args`. It initializes `num` to 12345 and `count` to 0. A `while` loop runs as long as `num` is greater than 0, dividing `num` by 10 and incrementing `count` in each iteration. Finally, it prints the value of `count` using `System.out.println(count);`. The terminal at the bottom shows the execution of `java DigitCount.java`, which outputs `5`.

```
1 public class DigitCount {
2
3     public static void main(String[] args) {
4         int num = 12345;
5         int count = 0;
6
7         while(num > 0)
8         {
9             num = num / 10;
10            count ++;
11        }
12        System.out.println(count);
13    }
14 }
```

Terminal Output:

```
PS D:\practice\lita\practice\anudip> javac Reverse.java
PS D:\practice\lita\practice\anudip> java Reverse
321
PS D:\practice\lita\practice\anudip> javac DigitCount.java
PS D:\practice\lita\practice\anudip> java DigitCount
5
PS D:\practice\lita\practice\anudip>
```

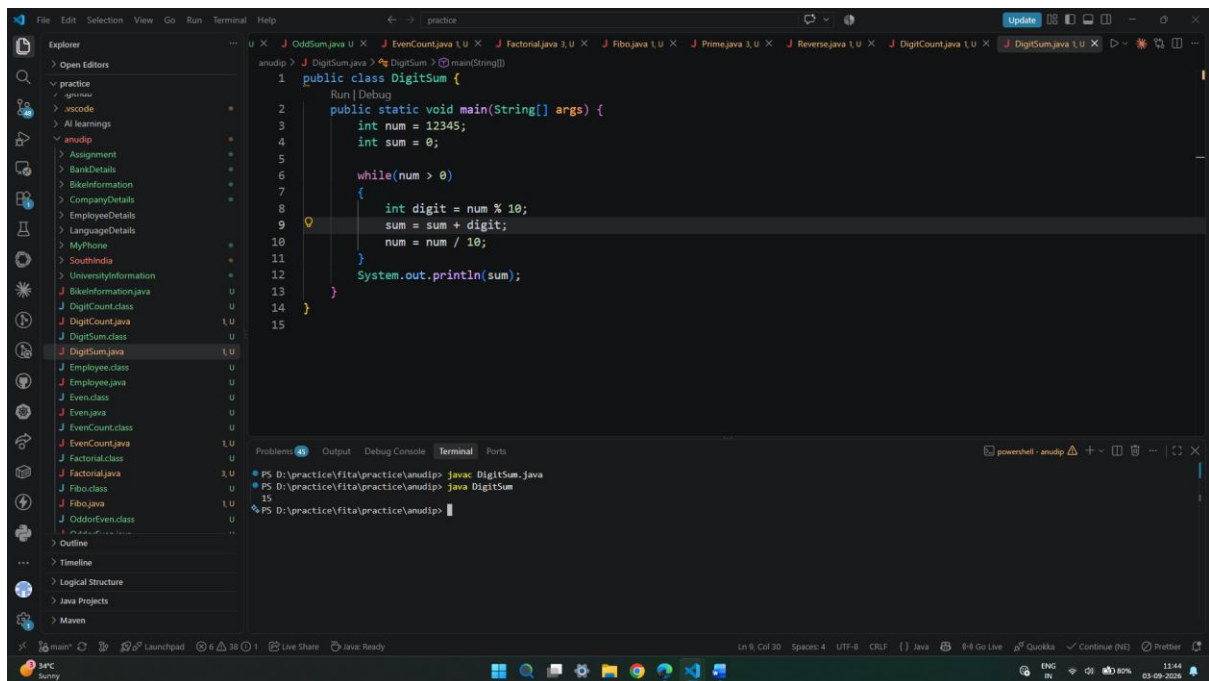
QUESTION 11:

Description: Sum of the number

Example:

Input = 123

Output = 6



The screenshot shows a code editor with a Java program named `DigitSum.java`. The program calculates the sum of the digits of a number 12345. The code is as follows:

```
1 public class DigitSum {
2     public static void main(String[] args) {
3         int num = 12345;
4         int sum = 0;
5
6         while(num > 0)
7         {
8             int digit = num % 10;
9             sum = sum + digit;
10            num = num / 10;
11        }
12        System.out.println(sum);
13    }
14 }
15
```

The terminal output shows the execution of the program:

```
PS D:\practice\lita\practice\anudip> javac DigitSum.java
PS D:\practice\lita\practice\anudip> java DigitSum
15
PS D:\practice\lita\practice\anudip>
```